Modelling and control of multibody mechanical systems

Model answers

Question 1

a) 1 degree of freedom, generalised coordinate is θ. [2 marks]

b) i) The position vector of the centre of mass of the first bar is

$$\boldsymbol{r_1} = \frac{l}{2}\left(\sin\theta\boldsymbol{i} + \cos\theta\boldsymbol{k}\right)$$

and the position vector of the centre of mass of the second bar is

$$\boldsymbol{r_2} = \frac{l}{2}\left(3\sin\theta\boldsymbol{i} + \cos\theta\boldsymbol{k}\right)$$

[3 marks]

ii) By differentiating each of the position vectors we obtain respectively the velocity vector of the centre of mass of the first bar as

$$\dot{\boldsymbol{r}}_{\boldsymbol{1}} = \frac{l}{2}\left(\dot\theta\cos\theta\boldsymbol{i} - \dot\theta\sin\theta\boldsymbol{k}\right)$$

and the velocity vector of the centre of mass of the second bar as

$$\dot{\boldsymbol{r}}_{\boldsymbol{2}} = \frac{l}{2}\left(3\dot\theta\cos\theta\boldsymbol{i} - \dot\theta\sin\theta\boldsymbol{k}\right)$$

Answers

[3 marks]

c) i) The total kinetic energy of the system is

$$T = \frac{1}{2}m\dot{\boldsymbol{r}}_{\boldsymbol{1}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{1}} + \frac{1}{2}m\dot{\boldsymbol{r}}_{\boldsymbol{2}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{2}} = \frac{1}{2}m\frac{l^2}{4}\dot\theta^2 + \frac{1}{2}m\frac{l^2}{4}\dot\theta^2\left(9\cos^2\theta + \sin^2\theta\right)$$

$$= ml^2\dot\theta^2\left(\frac{1}{4} + \cos^2\theta\right)$$

[3 marks]

ii) The potential energy of the system, with zero potential energy at the level of the point O, is

$$V = -mg\frac{l}{2}\cos\theta\times 2 = -mgl\cos\theta$$

[3 marks]

d) The Lagrangian function is

$$L = T - V = ml^2\dot\theta^2\left(\frac{1}{4} + \cos^2\theta\right) + mgl\cos\theta$$

The Lagrangian equation for the θ coordinate is

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot\theta}\right) - \frac{\partial L}{\partial\theta} = M$$

which yields

$$\frac{d}{dt}\left(2ml^2\dot{\theta}\left(\frac{1}{4}+\cos^2\theta\right)\right)+2ml^2\dot{\theta}^2\cos\theta\sin\theta+mgl\sin\theta=M$$

and finally gives

$$2ml^2\ddot{\theta}\left(\frac{1}{4}+\cos^2\theta\right)-2ml^2\dot{\theta}^2\sin\theta\cos\theta+mgl\sin\theta=M$$

[6 marks]

Answers

Question 2

a) 3 degrees of freedom, the generalised coordinates are ψ, θ and ϕ. [2 marks]

b) Three single-axis-rotation transformation matrices are needed.

$$D_\psi = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is the rotation matrix by angle ψ about a z axis.

$$C_\theta = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

which is the rotation matrix by angle θ about a y axis.

$$B_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

which is the rotation matrix by angle ϕ about an x axis. The angular velocity vector of the disc in arm-fixed coordinates is

$$\boldsymbol{\Omega} = C_\theta D_\psi \begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix} + C_\theta \begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + B_\phi^T \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot\psi\sin\theta + \dot\phi \\ \dot\theta \\ \dot\psi\cos\theta \end{bmatrix}$$

Answers

[3 marks]

c) The inertia matrix of the disc with respect to the (unspun) arm-fixed axes is

$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{zz} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

Direct application of the kinetic energy formula given in the notes, with the products of inertia set to zero, gives the kinetic energy for the rotation of the disc about its centre of mass as

$$T_r = \frac{1}{2} I_{xx} \left(-\dot\psi\sin\theta + \dot\phi\right)^2 + \frac{1}{2} I_{zz}\dot\theta^2 + \frac{1}{2} I_{zz}\dot\psi^2\cos^2\theta$$

The kinetic energy for the translation of the disc centre of mass is

$$T_t = \frac{1}{2} m v^2 = \frac{1}{2} m l^2 \dot\psi^2$$

in which $v = l\dot\psi$ is the magnitude of the velocity of the disc centre of mass. The kinetic energy of the system is therefore the sum of the two kinetic energies as

$$T = T_r + T_t = \frac{1}{2} I_{xx} \left(-\dot\psi\sin\theta + \dot\phi\right)^2 + \frac{1}{2} I_{zz}\dot\theta^2 + \frac{1}{2} I_{zz}\dot\psi^2\cos^2\theta + \frac{1}{2} m l^2 \dot\psi^2$$

[4 marks]

d) The height of the disc centre of mass doesn't change, therefore the potential energy of the system is $V = 0$. [2 marks]

e) Since the potential energy is zero, the Lagrangian function is

$$L = T = \frac{1}{2} I_{xx} \left(-\dot{\psi} \sin\theta + \dot{\phi} \right)^2 + \frac{1}{2} I_{zz} \dot{\theta}^2 + \frac{1}{2} I_{zz} \dot{\psi}^2 \cos^2\theta + \frac{1}{2} m l^2 \dot{\psi}^2$$

i) The Lagrangian equation for the ϕ Euler angle is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0$$

which yields

$$\frac{d}{dt} \left(I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \right) = 0 \tag{1}$$

and finally gives

$$I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) = C_0$$

in which C_0 is a constant.

ii) The Lagrangian equation for the θ Euler angle is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

which yields

$$\frac{d}{dt} \left(I_{zz} \dot{\theta} \right) + I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \dot{\psi} \cos\theta + I_{zz} \dot{\psi}^2 \cos\theta \sin\theta = 0$$

and finally gives

$$I_{zz} \ddot{\theta} + (I_{zz} - I_{xx}) \dot{\psi}^2 \sin\theta \cos\theta + I_{xx} \dot{\phi} \dot{\psi} \cos\theta = 0$$

iii) The Lagrangian equation for the ψ Euler angle is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\psi}} \right) - \frac{\partial L}{\partial \psi} = 0$$

which yields

$$\frac{d}{dt} \left(-I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \sin\theta + I_{zz} \dot{\psi} \cos^2\theta + m l^2 \dot{\psi} \right) = 0$$

and further gives

$$\frac{d}{dt} \left(-I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \right) \sin\theta - I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \dot{\theta} \cos\theta + I_{zz} \ddot{\psi} \cos^2\theta$$

$$-2 I_{zz} \dot{\psi} \dot{\theta} \sin\theta \cos\theta + m l^2 \ddot{\psi} = 0$$

and finally by using (1) it gives

$$-I_{xx} (-\dot{\psi} \sin\theta + \dot{\phi}) \dot{\theta} \cos\theta + I_{zz} \ddot{\psi} \cos^2\theta - 2 I_{zz} \dot{\psi} \dot{\theta} \sin\theta \cos\theta + m l^2 \ddot{\psi} = 0$$

[9 marks]

Answers

Question 3

a) Three single-axis-rotation transformation matrices are needed.

$$D_\psi = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is the rotation matrix by angle ψ about a z axis.

$$C_\theta = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

which is the rotation matrix by angle θ about a y axis.

$$B_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

which is the rotation matrix by angle ϕ about an x axis.

The transformation from Earth-fixed coordinates to arm-fixed coordinates is $\boldsymbol{A_1} = \boldsymbol{C_\theta D_\psi}$, the transformation from disc-fixed coordinates to arm-fixed coordinates is $\boldsymbol{A_2} = \boldsymbol{B_\phi^T}$, and the transformation from disc-fixed coordinates to Earth-fixed coordinates is $\boldsymbol{A_3} = \boldsymbol{D_\psi^T C_\theta^T B_\phi^T}$. [3 marks]

Answers

b) The angular momentum vector of the disc in arm-fixed coordinates is given by $\boldsymbol{H} = \boldsymbol{I\Omega}$, where $\boldsymbol{I}$ is the inertia matrix with respect to the disc-fixed axes (or arm-fixed axes since the radial moments of inertia of the disc are equal) and $\boldsymbol{\Omega}$ is the angular velocity vector of the disc in the arm-fixed coordinate system. To compute $\boldsymbol{H}$ in the arm-fixed coordinate system the transformations $\boldsymbol{A_1}$ and $\boldsymbol{A_2}$ are used. Hence $\boldsymbol{H}$ in arm-fixed coordinates is given by

$$\boldsymbol{H} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{zz} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \left(\boldsymbol{A_1} \begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix} + \begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + \boldsymbol{A_2} \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} \right) =$$

$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{zz} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \left(\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix} + \begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} \right) =$$

$$\begin{bmatrix} I_{xx}(-\dot\psi\sin\theta + \dot\phi) \\ I_{zz}\dot\theta \\ I_{zz}\dot\psi\cos\theta \end{bmatrix} \tag{2}$$

[4 marks]

c) By considering the rotational motion of the disc about its centre of mass O, according to $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$ evaluated using the arm-fixed coordinate system, one has

$$\frac{\mathrm{d}'\boldsymbol{H}}{\mathrm{d}t} + \boldsymbol{\Omega}_{\mathrm{arm}} \times \boldsymbol{H} = \boldsymbol{N}$$

where

$$\boldsymbol{\Omega}_{\text{arm}} = \boldsymbol{A_1}\begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot{\psi}\sin\theta \\ \dot{\theta} \\ \dot{\psi}\cos\theta \end{bmatrix}$$

is the angular velocity vector of the arm-fixed coordinate system and

$$\boldsymbol{N} = \begin{bmatrix} 0 \\ 0 \\ M \end{bmatrix}$$

where M is a constraint moment acting from the arm on the disc that prevents the disc from rotating relative to the arm about its radial axis perpendicular to the arm, thus reacting on the arm, and by consideration of the horizontal translation of the centre of mass of the disc, Newton's second law of motion gives (the horizontal acceleration of the disc centre of mass is $l\ddot{\psi}$)

$$\frac{-M\cos\theta}{l} = ml\ddot{\psi}$$

thus

$$M\cos\theta = -ml^2\ddot{\psi} \tag{3}$$

Therefore,

$$\begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi})\right) \\ I_{zz}\ddot{\theta} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{zz}\dot{\psi}\cos\theta\right) \end{bmatrix} + \begin{bmatrix} -\dot{\psi}\sin\theta \\ \dot{\theta} \\ \dot{\psi}\cos\theta \end{bmatrix} \times \begin{bmatrix} I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi}) \\ I_{zz}\dot{\theta} \\ I_{zz}\dot{\psi}\cos\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ M \end{bmatrix}$$

Answers

which yields

$$\begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi})\right) \\ I_{zz}\ddot{\theta} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{zz}\dot{\psi}\cos\theta\right) \end{bmatrix} + \begin{bmatrix} 0 \\ -I_{zz}\dot{\psi}\cos\theta(-\dot{\psi}\sin\theta) + I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi})\dot{\psi}\cos\theta \\ I_{zz}\dot{\theta}(-\dot{\psi}\sin\theta) + I_{xx}\dot{\theta}(-\dot{\psi}\sin\theta + \dot{\phi}) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ M \end{bmatrix} \tag{4}$$

The first equation of motion is therefore

$$I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi}) = C_0$$

in which C_0 is a constant, showing that the spin axis component of the angular momentum vector of the disc (see (2)) has a constant magnitude C_0 at all times. The second equation of motion is

$$I_{zz}\ddot{\theta} + (I_{zz} - I_{xx})\dot{\psi}^2\sin\theta\cos\theta + I_{xx}\dot{\phi}\dot{\psi}\cos\theta = 0$$

The third equation of motion is found by multiplying the third row in (4) by $\cos\theta$ and substituting from (3), which gives

$$-I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi})\dot{\theta}\cos\theta + I_{zz}\ddot{\psi}\cos^2\theta - 2I_{zz}\dot{\psi}\dot{\theta}\sin\theta\cos\theta + ml^2\ddot{\psi} = 0$$

[8 marks]

d) i) The second equation of motion now has the moment M_y on the right hand side, therefore

$$M_y = I_{zz}\ddot{\theta} + (I_{zz} - I_{xx})\dot{\psi}^2 \sin\theta\cos\theta + I_{xx}\dot{\phi}\dot{\psi}\cos\theta$$

and therefore

$$M_y|_{\theta=\theta_t} = (I_{zz} - I_{xx})\dot{\psi}^2 \sin\theta_t\cos\theta_t + I_{xx}\dot{\phi}\dot{\psi}\cos\theta_t$$

[2 marks]

ii) For the initial motion of the disc ($\dot{\psi} = 0$, $\theta = 0$, $\dot{\phi} = \omega_{spin}$), it can be concluded from the first equation of motion that

$$I_{xx}\omega_{spin} = C_0 \tag{5}$$

For the final motion (while the spin axis of the disc is maintained at $\theta = \theta_t$), from the third equation of motion it can be determined that $\ddot{\psi} = 0$ since $\dot{\theta} = 0$, therefore $\dot{\psi}$ is constant. From the first equation of motion it can then be concluded that $\dot{\phi}$ will also remain constant at some value, say ω_{spinf}. The first equation of motion therefore gives

$$I_{xx}(-\dot{\psi}\sin\theta_t + \omega_{spinf}) = C_0$$

in which the magnitude of the spin axis component of the angular momentum vector of the disc remains constant at C_0. This equation together with (5), can be used to provide the value of $\dot{\psi}$ as

$$\dot{\psi} = \frac{\omega_{spinf} - \omega_{spin}}{\sin\theta_t}$$

Answers

[3 marks]

Question 4

a) By differentiating the velocity vector of the centre of mass of the first bar we obtain its acceleration vector as follows

$$\ddot{\boldsymbol{r}}_{\mathbf{1}} = \frac{l}{2}\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\boldsymbol{i} - \frac{l}{2}\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\boldsymbol{k}$$

and by differentiating the velocity vector of the centre of mass of the second bar we obtain its acceleration vector as follows

$$\ddot{\boldsymbol{r}}_{\mathbf{2}} = \frac{3l}{2}\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\boldsymbol{i} - \frac{l}{2}\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\boldsymbol{k}$$

[3 marks]

b) The total force vector acting on the first bar is

$$\boldsymbol{F}_{\mathbf{1}} = F_{1x}\boldsymbol{i} + F_{1z}\boldsymbol{k} + mg\boldsymbol{k} - F_{2x}\boldsymbol{i} - F_{2z}\boldsymbol{k}$$

and the total force vector acting on the second bar is

$$\boldsymbol{F}_{\mathbf{2}} = F_{2x}\boldsymbol{i} + F_{2z}\boldsymbol{k} + mg\boldsymbol{k} + F_R\boldsymbol{k}$$

where F_{1x} and F_{1z} are the horizontal and vertical components, respectively, of the constraint force from Earth acting on the first bar at point O, F_{2x} and F_{2z} are the horizontal and vertical components, respectively, of the constraint force holding the two bars together at point A, and F_R is the vertical force of constraint applied by the slider on the second bar at point B. Note that the horizontal force applied by the slider on the second bar at point B is zero, because the slider is massless and its horizontal motion is not restricted by friction. [3 marks]

Answers

c) i) Newton's second law of motion applied on the second bar gives

$$\boldsymbol{F}_{\mathbf{2}} = m\ddot{\boldsymbol{r}}_{\mathbf{2}} \tag{6}$$

and therefore equating the coefficients of $\boldsymbol{i}$ yields the horizontal component of the constraint force acting on the second bar at point A as

$$F_{2x} = \frac{3ml}{2}\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right) \tag{7}$$

[3 marks]

ii) Newton's second law of motion applied on the first bar gives

$$\boldsymbol{F}_{\mathbf{1}} = m\ddot{\boldsymbol{r}}_{\mathbf{1}} \tag{8}$$

and therefore equating the coefficients of $\boldsymbol{i}$ yields

$$F_{1x} - F_{2x} = \frac{ml}{2}\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)$$

and by substitution from (7), gives the horizontal component of the constraint force acting on the first bar at point O as

$$F_{1x} = 2ml\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right) \tag{9}$$

[3 marks]

d) By equating the coefficients of $\boldsymbol{k}$ in (8) gives

$$F_{1z} - F_{2z} + mg = -\frac{ml}{2}\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right) \tag{10}$$

Also by considering the rotational motion about the centre of mass of the first bar and since its moment of inertia is negligible, gives

$$-F_{1x}\frac{l}{2}\cos\theta + F_{1z}\frac{l}{2}\sin\theta - F_{2x}\frac{l}{2}\cos\theta + F_{2z}\frac{l}{2}\sin\theta + M = 0$$

and after substitution from (7) and (9), yields

$$(F_{1z} + F_{2z})\sin\theta - \frac{7ml}{2}\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\cos\theta + \frac{2M}{l} = 0 \tag{11}$$

Equations (10) and (11) can be solved simultaneously to give

$$F_{1z}\sin\theta = ml\left(-\frac{\ddot{\theta}}{4} + 2\ddot{\theta}\cos^2\theta - 2\dot{\theta}^2\sin\theta\cos\theta\right) - \frac{mg\sin\theta}{2} - \frac{M}{l} \tag{12}$$

and

$$F_{2z}\sin\theta = ml\left(\frac{\ddot{\theta}}{4} + \frac{3}{2}\ddot{\theta}\cos^2\theta - \frac{3}{2}\dot{\theta}^2\sin\theta\cos\theta\right) + \frac{mg\sin\theta}{2} - \frac{M}{l} \tag{13}$$

By equating the coefficients of $\boldsymbol{k}$ in (6) gives

$$F_{2z} + mg + F_R = -\frac{ml}{2}\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right) \tag{14}$$

and by multiplying both sides of this equation by $\sin\theta$ and also substituting from (13) yields

$$F_R\sin\theta = -ml\left(\frac{3\ddot{\theta}}{4} + \ddot{\theta}\cos^2\theta - \dot{\theta}^2\sin\theta\cos\theta\right) - \frac{3}{2}mg\sin\theta + \frac{M}{l} \tag{15}$$

Furthermore, by considering the rotational motion about the centre of mass of the second bar and since its moment of inertia is negligible, gives

$$F_{2x}\frac{l}{2}\cos\theta + F_{2z}\frac{l}{2}\sin\theta - F_R\frac{l}{2}\sin\theta = 0$$

and by substituting from (7), (13), and (15) finally yields the equation of motion of the slider-crank system as

$$2ml^2\ddot{\theta}\left(\frac{1}{4} + \cos^2\theta\right) - 2ml^2\dot{\theta}^2\sin\theta\cos\theta + mgl\sin\theta = M$$

[8 marks]

Answers